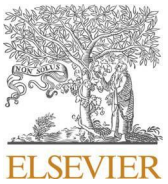
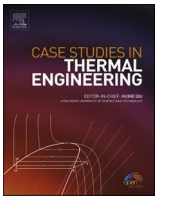
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Deep learning-based approach to R-134a bubble detection and analysis for geothermal applications

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ABSTRACT

This study utilized deep learning to analyse bubble images of R134a fluid produced on bare and novel oleophilic coated stainless steel heat exchanger plates to compare the heat transfer performance in terms of bubble parameters. This work was carried out as a part of the GeoHex project. The vapour bubbles of the boiling R134a fluid, produced on two types of heat exchanger plates, were captured with a high-speed camera during a boiling experiment. A deep learning algorithm based on the Mask *R*–CNN model was trained for bubble detection in the experimental videos. The algorithm achieved a bubble detection accuracy of 61.22% and 78.6% for the coated and uncoated plates, respectively. The lower accuracy for the coated plate could be attributed to the challenge of distinguishing bubbles from the complex coating pattern. The algorithm determined the number of detected bubbles, bubble sizes, and centroids per frame, tracking each bubble frame by frame. The tracking results were used to calculate the bubble parameters such as departure frequency, departure diameter, and active nucleation site density. The calculated departure diameter for both plates exhibited a difference of 3%, which corresponded closely to the experimental results showing a 4% variation in heat transfer coefficient, which was measured under identical experimental conditions utilizing the GeoHex rig. However, the calculated departure frequency and active nucleation site density exhibited a 17% disparity between the coated and uncoated plates, primarily stemming from the lower detection accuracy for the coated plate. Some limitations of the trained model were identified in accurately detecting very small, out of focus, and mutually occluded bubbles. Future research should focus on improving experimental rig designs to capture better side-view images and develop advanced machine learning or other techniques to address these limitations.

1. Introduction

Geothermal energy has the potential for several applications, including power generation, geo-exchange, and direct thermal application. In addition, geothermal energy is renewable (when reasonably exploited) and has a low carbon impact compared to fossil fuel use; hence its increased use will lead to reduced use of fossil fuels in power generation and mitigate global warming. However, as can be seen from Fig. 1, the conversion efficiency of geothermal power developments is generally lower than that of conventional thermal power plants [3]. Therefore, this conversion efficiency should be taken into account when calculating the power potential of

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newly drilled geothermal wells and for resource estimation studies. A study suggests that the power conversion efficiency from geothermal resources ranges from 10 to 17% [4]. One of the significant factors beyond the thermodynamic limitation responsible for this low efficiency is a lower limit of reinjection temperatures to avoid silica scaling as the solubility of silica decreases with temperature; hence, the full potential of geothermal energy cannot be exploited. Another factor is the relatively low temperature of the geothermal heat source (compared to combustion of fuel), which is a physical limit beyond technical solutions.

The GeoHex project addressed the scaling and heat transfer performance of heat exchangers used in geothermal power plants. One of the GeoHex project's aims is to analyse heat transfer through nucleate boiling in an evaporator to enhance the heat transfer performance of phase change heat exchangers since the heat transfer performance of nucleate boiling is several times higher than that of conventional film boiling. The GeoHex project is developing high-performance heat exchanger (HX) coating materials to promote nucleate boiling (for ORC working fluid) induced by the GeoHex-engineered surface for geothermal applications, mainly in binary/ORC power plants. During nucleate boiling, bubbles are formed because vapour molecules become trapped in surface defects or imperfections; such locations where the vapour bubbles are forming are known as nucleation sites [5]. Therefore, Surface modification techniques, such as altering surface roughness using sandpaper treatment, creating artificial cavities, and applying surface coatings, have been employed to enhance nucleate boiling and improve heat transfer performance; these techniques aim to generate surface structures that facilitate the formation and expansion of vapour bubbles during boiling, this promotes various advantages, including reduced boiling incipient superheat and an enhanced critical heat flux limit [6–8]. Surface roughness enhances heat transfer by increasing surface area, which tends to increase the number of nucleation sites, thereby generating a greater number of bubbles [9]. It also shows in reference [10] that critical heat flux is delayed by increased surface roughness [11]. performed thermal performance optimization of heat pipes using CuO–H2O nanofluid; after boiling nanofluid, they found both the contact angle and surface roughness of copper pipe were increased due to the deposition of nanoparticles on copper substrate. That means due to the deposition of nanofluid on the copper substrate, surface roughness increases, resulting in the increased value of critical heat flux and heat performances of the heat pipe. The deposition of nanoparticles causes the reduction of contact angle, which increases wettability and thermal performance [11].

GeoHex project partner developed oleophilic (the oleophilic surface that consisted of a hydrophilic surface which was then dotted with a pattern of islands of hydrophobic material) coated plate to evaluate the heat transfer performance with respect to heat transfer coefficient and bubble parameters because the Hydrophilic surfaces, with static contact angles (CA) of $<90^{\circ}$, as well as hydrophobic surfaces, with CA of $>150^{\circ}$ can increase heat transfer, this because there are competing influences related to increased bubble nucleation density with hydrophobic surfaces, but also lack wicking and rewetting [12].

Before manufacturing a prototype GeoHex ORC system, appropriate coating materials are selected that may improve heat transfer efficiency. Each selected GeoHex coating was developed on a 10×10 cm^2 stainless steel substrate. The coated/uncoated plate was then mounted within a heat exchanger in a GeoHex test rig to evaluate the selected coating material's heat transfer performance. The R134a fluid flows on the coated side, and hot water flows on the other side of the coated plate. An uncoated stainless steel heat exchanger plate was evaluated similarly for comparison.

In this work, bubble images were captured using a high-speed camera and analysed using a deep learning algorithm to calculate parameters such as bubble departure diameter, departure frequency, active nucleation site density, etc. These data are beneficial for validating two-phase flow and heat transfer models, including computational fluid dynamics with interface tracking methods. The number of nucleation sites per unit area of the heater surface, which is the active nucleation site density (NSD), is one of the critical parameters used in modelling and predicting nucleate boiling heat transfer. The nucleation site density and other bubble parameters evaluated in this work will be used in the simulation of bubble dynamics which is being developed by another partner in the GeoHex project.

The deep learning algorithm developed in this work can also analyse other bubbles, such as water and thermal oil bubbles. Deep learning has recently been recognised as a powerful tool in digital image processing [13–16]. Deep learning is a specialised subset of machine learning, a subset of artificial intelligence (AI).

2. Experimental setup

In this work, R134a was used as a working fluid. The boiling temperature of R134a is $-26.3\ ^{\circ}$C at atmospheric pressure. The boiling

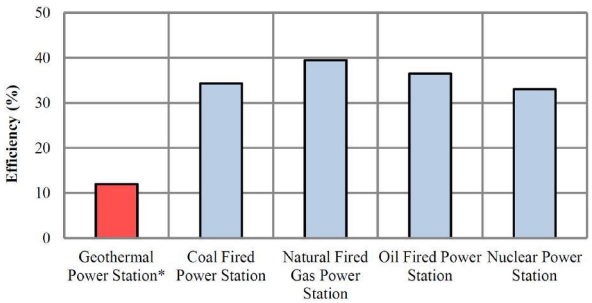


Fig. 1. Efficiency comparison of different power plants [1,2] [3];

temperature of R134a was set at 35 °C, at which the pressure is 9 bar. The maximum temperature and pressure are the set point of safety device intervention. The design parameters are higher for safety reasons. The heat transfer fluid hot water temperature was set to 80 °C to increase the amount of boiling fluid and to allow a flow rate of 0.38 kg/min.

Test rig designed parameters are listed in Table 1 with precision.

The boiling heat exchanger and a test rig are shown in Fig. 2, developed by Spike Renewables S. r.l, a GeoHex project partner. The piping and instrumentation diagram (P&ID (Fig. 3), shows the position of all components and sensors. All the sensors are connected to a programmable logic controller (PLC), which is responsible for the test rig control (start/stop) and the data acquisition. The heat exchanger has been developed with a two-sided optical window to capture bubble video footage during a boiling experiment (Fig. 4).

2.1. Boiling experiment

The video footage of the bubbles was captured during boiling experiments at the University of Iceland (UoI). For the boiling experiment during bubble image capture, very low heat flux was used to heat the heat exchanger surfaces to have a less intense boiling experiment; since the bubble images are aiming to be analysed by tracking individual bubbles, a simplistic bubble video with fewer bubbles would be easier to analyse. Table 2 lists the experimental parameters used for the bubble image capture experiments.

Enormous numbers of bubbles generate during the boiling phase of a liquid; therefore, it becomes very challenging to analyse the bubbles, track individual bubbles, and find the nucleation sites for each bubble. Hence, removable arrangements were made in the test rig to capture video of the bubbles from the top glass window (top view) and the side glass window (side view). For the top view, a glass tube (with both ends open) of diameter 2.0 cm was placed on the ORC side of the plate, as shown in Fig. 5. This setup allows fewer bubbles to generate during the boiling phase and makes it possible to perceive the bubbles below the liquid surface. Therefore, the video footages were almost straightforward to analyse, and lower heat flux was also used to boil the R134a fluid. For the side view, a vertical plate was placed on top of the coated plate, as shown in Fig. 5, and this setup allows to focus on fewer bubbles while the image capture is taking place from the side window.

The bubble images were captured using a high-speed video camera *Phantom VEO-E 310L;* for better lighting, an LED ring light (with 150 LEDs in the rings) was used, as shown in Fig. 6.

2.2. The camera setup used for the boiling experiments was as follows

2.2.1. Top view

The top view in this study refers to the perspective from which the bubble videos were captured as the bubbles rose upwards. This view provides a clear focus on the bubbles during their ascent, allowing for analysis of their characteristics and behaviours.

The top view was captured with the following configurations.

- 2nd smallest aperture was used to have a better depth of field.
- Resolutions: 768 × 768
- Frame Per Second (FPS): 600

The top view, while providing a comprehensive view of the rising bubbles, faces limitations in accurately capturing very small bubbles generated during boiling. These small bubbles, located further down and often out of focus, present a challenge in precisely determining their characteristics and calculating relevant bubble parameters.

2.2.2. Side view

The side view should offer a valuable perspective for capturing bubble dynamics. It allows for a more accurate calculation of parameters such as departure diameter and departure frequency, as the bubbles remain in focus throughout their rise without significant sideways movement. The following were the conditions during capturing the side view:

Nothing was visible for the 2nd smallest aperture; therefore, the 2nd largest aperture was used, which allows light and bubbles to be better visible but is still not as good as the bubbles seen through the top view. Due to the use of a larger aperture, the depth of the field was inferior. Therefore, most bubbles were out of focus.

The experimental setup to capture bubble images from the top viewing port (top view) and the side viewing port (side view) is shown in Fig. 7.

Although the side view offers advantages in accurately calculating parameters such as departure diameter and departure frequency, it was not extensively utilized in this study due to the inferior focus and low quality of the side view bubble video. This happens due to the constraints in the rig design and experimental setup. Accessing the side view in practical scenarios can be challenging, limiting its widespread application in bubble analysis studies.

2.2.3. Top view vs side view

The top view remains crucial in bubble analysis studies despite its limitations. While the side view provides valuable insights,

Table 1
Boiling experiment rig design parameters.

Property	Unit	Range	Precision
Temperature	^{0}C	20–80	0.1 °C
Pressure	bar	0–9	0.1 bar
Flowrate	L/min	0–3.2	0.1 L/min



Fig. 2. Boiling experiment rig.

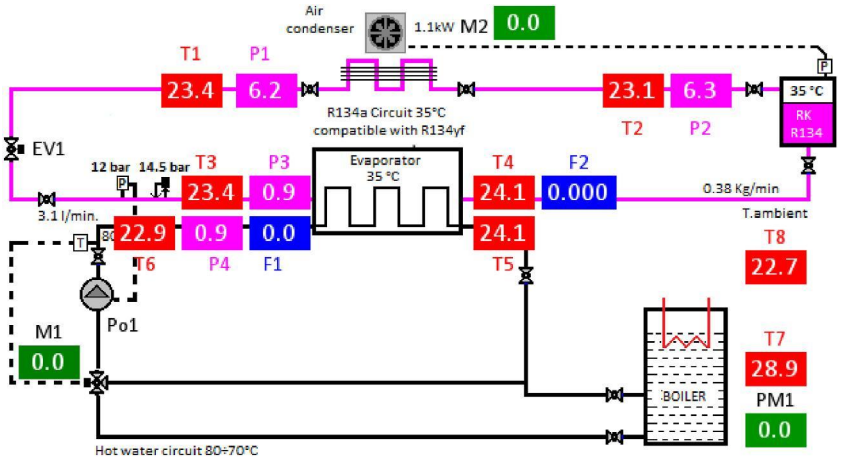


Fig. 3. Schematic of the flow loop of the two fluids, indicating the different measurement points (P&ID).



Fig. 4. To capture images, the heat exchanger shows the optical light windows (top and side).

gaining access to this perspective was challenging in our experiment. This could also be true for many real-world conditions. As a result, the top view becomes indispensable as it offers a more accessible and practical approach for capturing and analyzing bubble dynamics. By focusing on the top view, this study provides valuable findings that can be widely applied in various scenarios where the side view is difficult to obtain.

Table 2
Boiling experiment parameters during bubble image capture.

Heat exchanger plate	Substrate Material	Coating Material	Area, cm^2	Pressure, bar	Flow rate, kg/min	LMTD, ^0C	Heat Flux, kW/m^2
Stainless Steel Plate	Stainless Steel	None	10 × 10	9	0.38	15	18
Coated Plate	Stainless Steel	Oleophilic Coating -Hydrophobic dots (Si:Nd:Zr-57:26:17 at%) over hydrophilic (Si:La:Zr – 60:20:20 at%) material	10 × 10	9	0.38	15	18

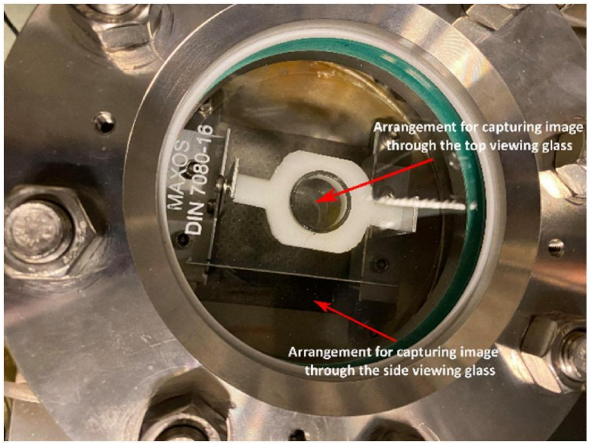


Fig. 5. Experimental setup within the heat exchanger to capture bubble videos from the top and side viewing windows.



Fig. 6. Phantom VEO-E 310 L camera with attached LED light (left) LED ring light assembly (right).

For both the top and side views, the mean temperature difference (LMTD) between water and R134a was 15 °C for all the experiments, and the corresponding heat flux was about 18 kWm^{-2} measured in the test rig.

Two types of plates were studied to capture bubble images for the boiling experiments.

1. Uncoated Stainless-steel plate (316 L) (left image in Fig. 8).
2. Coated stainless steel plate – Oleophilic (*Hydrophobic dots [Si:Zr:Nd] over hydrophilic [Si:Zr:La] surface)*

Coating on the stainless-steel substrate (316 L) on the ORC side (right image in Fig. 8).

Fig. 9 shows the bubbles produced in the coated and uncoated stainless-steel plates for the top view. The side view video was only recorded for the coated plate and is shown in Fig. 10.

3. Methodology

The bubble videos were captured in the test rig at UoI during a typical evaporation/boiling experiment. Using the Phantom Camera Control (PCC) software, the original video was converted to an MP4 file. Video qualities also improved to a level of better/higher visibilities allowing more light using this software. The converted videos were then analysed using the code based on the deep learning algorithm.

The following steps of the deep learning algorithm are used to develop the code to detect and track the individual bubbles per frame and find their size distribution. Those are then used to calculate the various bubble parameters. The same methodology was used for

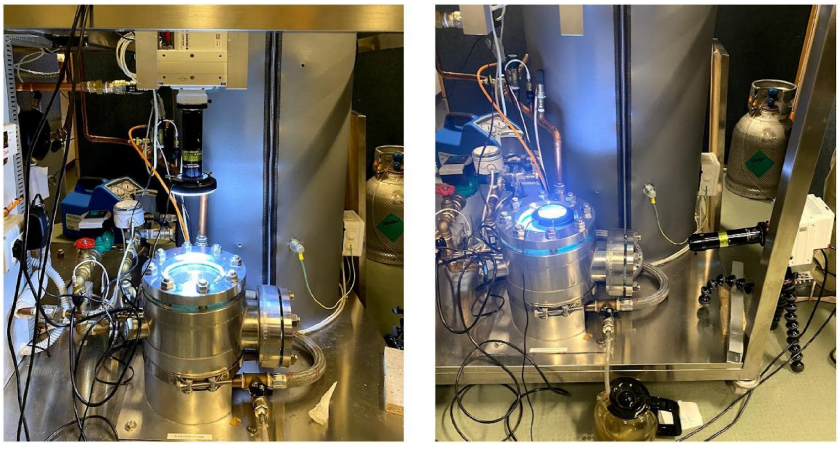


Fig. 7. Experimental setup to capture bubble videos; top view (left), side view (right).



Fig. 8. Uncoated (left) and coated (right) heat exchanger plates.

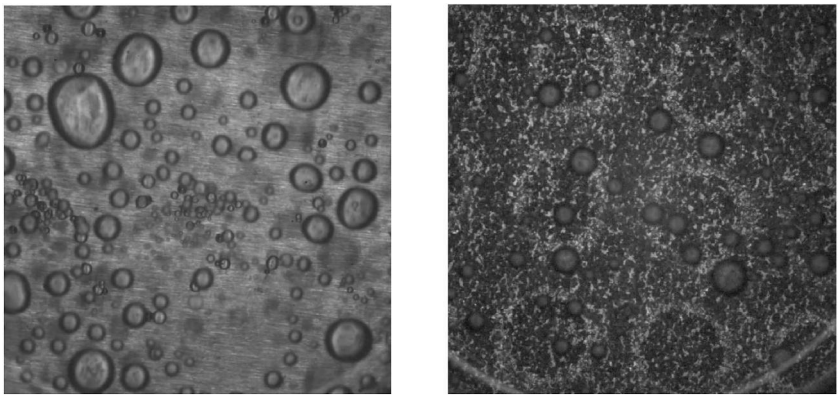


Fig. 9. Bubbles generated on the Uncoated plate (left) and Coated plate (right) - Top view.

both the coated and uncoated plates. For example, the various processes for the uncoated stainless-steel plate are illustrated and explained below.

3.1. Image processing algorithm

The algorithm is divided into three parts.

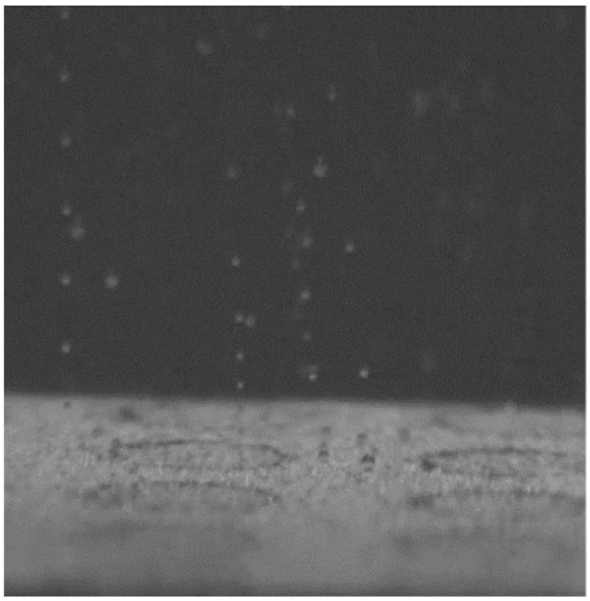


Fig. 10. Bubbles generated on the Coated plate - Side view.

I. **Pre-processing**: The image or video footage captured by the high-speed camera is used as input. The algorithm aims to prepare the images using specific and deep learning methods to detect the bubbles and prepare the image to perform some analytics.
II. **Training, testing and validation of model**: In this step, we trained a deep learning model with the bubble videos to detect the bubbles. We tested the model and validated it as well.
III. **Post-processing algorithm**: The output image from training, testing and validation of the model is used as input for the post-processing algorithm. In this part of the algorithm, the individual bubbles are tracked frame by frame, calculating the essential numerical data or properties relevant to bubble dynamics. Fig. 11 shows the deep learning image processing algorithm with pre-processing, training, testing and validation of the model and post-processing steps.

3.1.1. Pre-processing

We prepare the image dataset at the very beginning to train and test the model. The following steps were carried out in this part of

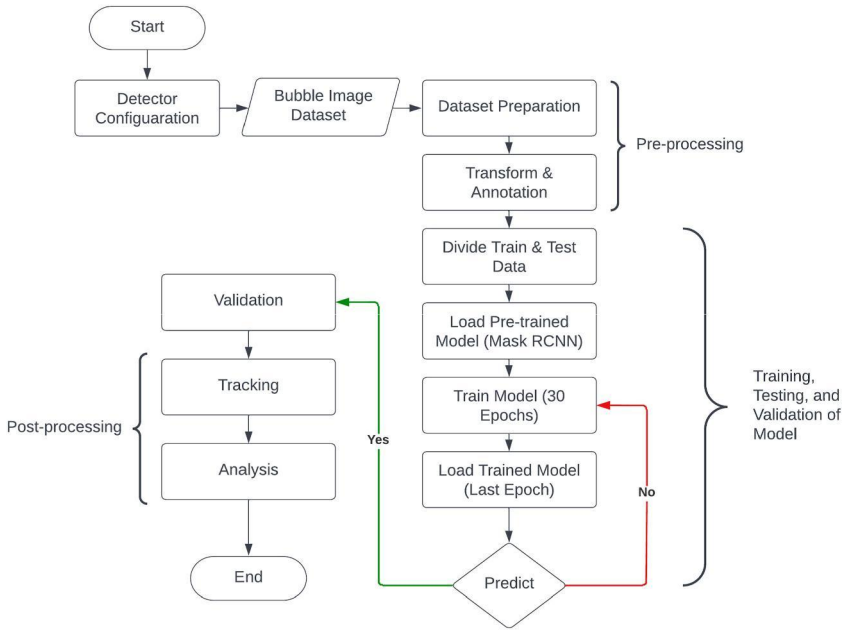


Fig. 11. Deep learning image processing algorithm.

the algorithm.

3.1.1.1. Detector configuration. Configuration for training bubble detection on the boiling images or video dataset: The GPU count is 1, images per GPU is 8, and backbone is resnet 50, minimum and maximum image dimension is 256 and 256, train ROIs per image is 32 and finally, steps per epoch is 25.

3.1.1.2. Detector dataset. Operation of a dataset for training bubble detection on the boiling image or video dataset, such as labelling, loading dataset with unique ids, image height and width, area of bubbles in images and creating objects for every image in the dataset with its features, draw the shape and load mask of each bubble in each image.

3.1.1.3. Datasets preparation. This stage annotates the image dataset, and then the boiling dataset is prepared using the detector dataset's operation. The image before and after an annotation is shown in Fig. 12.

3.1.1.4. Transform annotations. After dataset preparation, we configure the annotation for our convenience, i.e., parse each annotation individually and take each point into a polygon list of bubble area, generating unique ids, labels, and objects of the image. The transformed annotations are shown in Fig. 13.

3.1.2. Training, testing, and validation of the model

To detect the bubbles, we trained the model independently for coated and uncoated video cases as they differ significantly in colour and feature. We sampled 20 images from each video dataset and separated them into 15 and 5 images for training and testing. The Mask *R*–CNN model was then trained on 15 images, and its performance was assessed on 5 test images for both scenarios. We used the Mask *R*–CNN model as it is a conceptually simple, flexible, and general framework for object instance segmentation [17]. Finally, we validated the model by detecting the bubbles, as seen in Fig. 14.

The following steps were carried out to perform this part of the algorithm.

3.1.2.1. Divide train/test data. We sampled 20 unique frames from the video, divided 20 images into 15 images for training and 5 for testing, and then ran the operation of the detector dataset.

3.1.2.2. Load pre-trained model. We downloaded pre-trained MS COCO weights that implement Mask *R*–CNN [17] on Python 3, Keras and Tensorflow. The model generates bounding boxes and segmentation masks for each instance of an object in the image. It's based on Feature Pyramid Network (FPN) and a ResNet 101 backbone.

3.1.2.3. Train model (30 epochs). We set up the configuration for training the model using detector configuration and loaded pre-train weights where the learning rate is 0.001, and the epochs are 30.

3.1.2.4. Load trained up model (last epoch). After completing the training, the last trained weights are loaded as the last epoch is the most efficient.

3.1.2.5. Prediction/validation. In this step, run the experiment video of boiling flow through the model to show the performance of the model. The bubbles are detected in each frame of the video (Fig. 14).

To show that the code can detect similar objects in any image, for example, an image of bubbles [18] was analysed using this code. Fig. 15 shows an example where the developed deep learning code could detect about 99% of the object in that image.

3.1.3. Post-processing

In this part of the algorithm, the bubbles were tracked frame by frame and analysis was performed.

3.1.3.1. Tracking and analysis. Centroids, radii and the number of bubbles were detected in each frame. In this stage, OpenCV was used to analyse further and annotate circles around the detected bubbles in each frame of the experimental videos. A minimum Euclidean distance was found by trial and error to identify and track the same bubble in subsequent frames. Then an object was constructed for each video frame containing the number of bubbles, along with the bubble centre coordinates (x, y) and radii (r). The trajectory of each bubble throughout all the frames was then stored in a data frame.

The analysis was then performed by plotting the bubble diameter distribution, the number of bubbles with time variation, and the variation of diameters of a single bubble with time and calculating the nucleation site using the K-means algorithm. The data frame was used to group the bubbles using the K-means clustering algorithm, an unsupervised machine learning algorithm to calculate nucleation

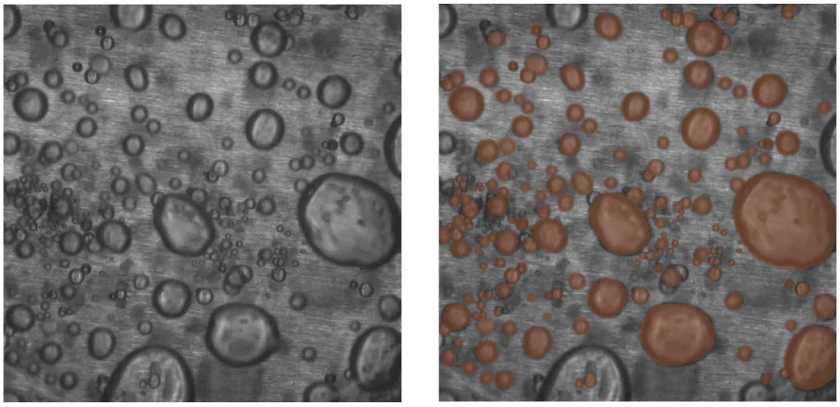


Fig. 12. Annotated image (right), an image before annotation (left).

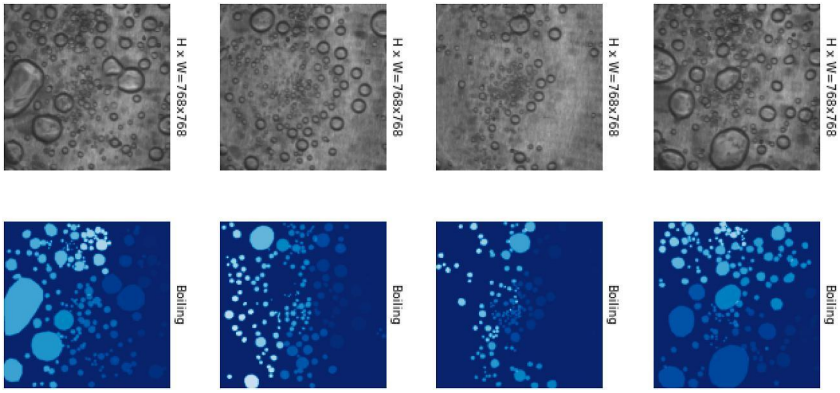


Fig. 13. Transformation of annotated images: Original images (top) transformed images (bottom).

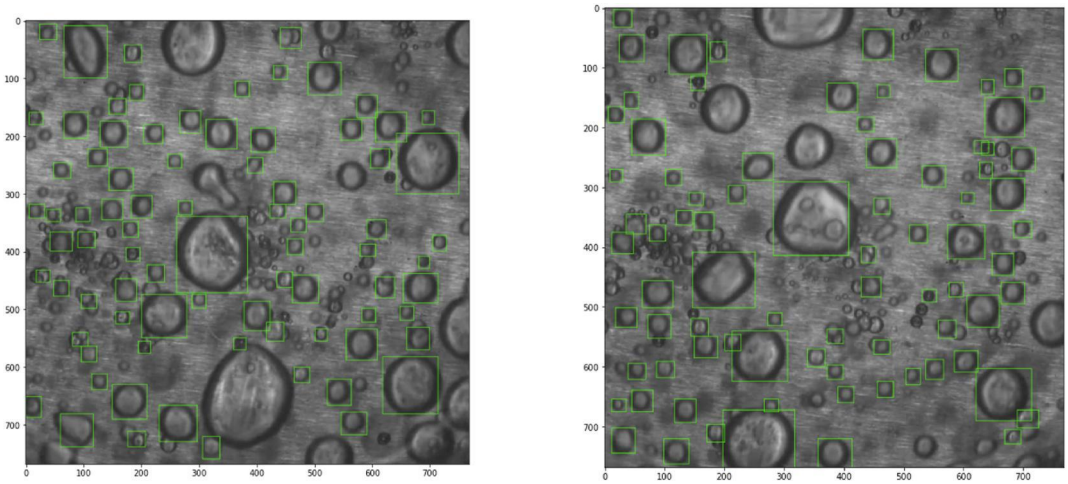


Fig. 14. Bubble detection in the test images.

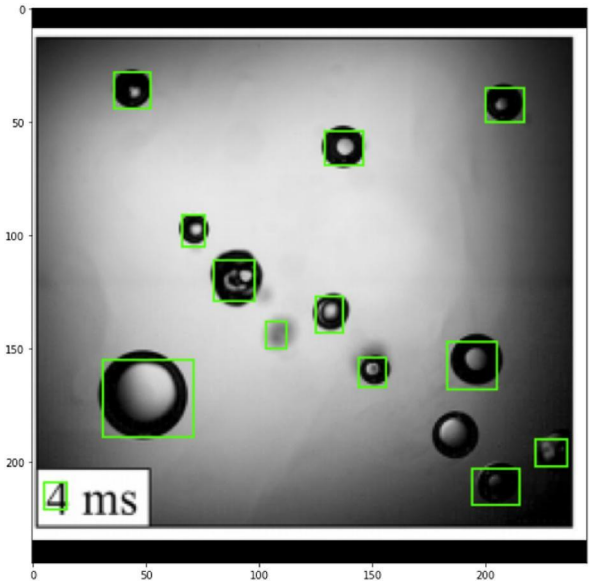


Fig. 15. Image of blood cells, the deep learning code can detect almost 99% of the cells in this image [18].

sites. The optimum group number (K) was found after 150 iterations. It was determined that K = 50 replicated the nucleation sites with the most accuracy for the un-coated plate and K = 42 for the coated plate. After that, departure diameter, departure frequency and active nucleation site densities are calculated using the bubble tracking data and nucleation sites. Fig. 16 shows the nucleation site with ' × ' sign evaluated by the K-means clustering method for the uncoated plate.

4. Results and Discussions

For a coated plate evaluation, both the top view and side view videos were recorded; for the uncoated stainless-steel plate, only the top view videos were recorded. Because of the geometry of the side viewing port of the heat exchanger, LED lights were not sufficient to illuminate the heat exchanger plate, a large flashlight was also tried, but the recorded image was still quite dark. That's why the larger aperture was used to allow plenty of light. However, this also reduced the depth of field. Therefore, the bubble image recorded through the side viewing glass was not as good as those recorded from the top viewing glass and, therefore, not analysed. While recording the bubble videos from the top viewing glass, 2nd smallest aperture and LED ring lights were used, and the bubble videos looked excellent. Nevertheless, the videos still required some processing before analysis, especially since more lights were needed in the videos to improve the bubbles' visibility, which was done using the PCC software.

In the post-processing phase, to characterise the bubble behaviours, the deep learning code tracked each bubble per frame from where they appeared until they collapsed. The deep learning code calculated the number of bubbles and their sizes per frame, the centroid of each bubble, bubble size distribution, and nucleation sites. Active nucleation site density, bubble departure diameter, and bubble departure frequency were then calculated using the post-processing analysis results and using equations (1)–(4) in the following subsections.

4.1. Bubble size, distribution, and accuracy

The number of bubbles with time and the variation of one bubble diameter with time, respectively, are shown in Figs. 17 and 18 for the coated and uncoated plates; Figs. 17 and 18 show the number of bubbles and the diameter of a bubble varies with time. This variation occurs because bubbles are born on the surface, then grow (diameter increases), the coalescence breaks up or collapses, and new bubbles form over time.

Plate.

The variation of the diameter of a bubble with time is not very well understood [19]. developed a bubble diameter model. According to this model, the bubble size can be divided into three parts according to the bubble transformation manner after its generation. The bubble diameter d_b is defined as follows:

$$d_b = f\left(d_d, d_{heat}, d_{collision}, V\right) \tag{1}$$

where, d_d is the bubble departure diameter taken as the initial diameter, d_{heat} is the bubble size variation contribution due to the heat transfer, $d_{collision}$ is the bubble size variation contribution due to collisions, and V is the separating volume function representing the bubble size variation due to the phase interaction force. The collision can lead to coalescence or break up; this is shown in Fig. 19.

Note that all the events shown in Fig. 19 can happen when boiling R134a. Fig. 20 shows a histogram of the bubble diameter of the coated and uncoated plates. According to the shape of the histogram (Fig. 20), the diameter distributions of bubbles are very consistent with the log-normal distribution. A Log-Normal distribution is a continuous probability distribution of a random variable whose logarithm is normally distributed. Unlike a standard normal distribution, the log-normal distribution does not follow a bell curve, is not symmetrical around the peak, and has a significant positive skew in one direction. If a distribution shows a normal distribution after taking $\log_{10}$ in the x-scale of this distribution, then this distribution is a log-normal distribution. Fig. 21 shows the distribution after converting the linear x-scale to a $\log_{10}$ scale, and the distribution is now almost symmetrical around the peak. The fitted curves in Fig. 21 indicate a normal distribution of bubble diameters. Therefore, we can say that R134a fluid bubble diameter distribution is a log-normal distribution, as shown in Fig. 20. Others saw this log-normal bubble diameter distribution (Hart, 2008 [20]; for the water bubbles; log-normal behaviour was also observed in the lubricant bubble distribution [21]. However, no results on R134a have been found in the literature.

The bubble detection accuracy was calculated to evaluate the efficiency of the deep learning code. To find the bubble detection accuracy, we considered several random frames. Then, the bubbles were counted and numbered; those were detected by code and eyes. Figs. 22 and 23 show the marked bubbles detected by the code for the coated and uncoated plates and the visual counting. The accuracy of bubble detection for the coated and uncoated plates is 61.22% and 78.6%, respectively. Our algorithm has certain limitations when it comes to detecting very small bubbles and addressing the mutual occlusion of bubbles. We acknowledge that the segmentation ability of our Mask RCNN-based model is not optimal for accurately detecting and segmenting tiny bubbles. This limitation results in some errors when calculating bubble departure diameters, as the actual departure diameters may be smaller than the calculated values due to undetected tiny bubbles.

Additionally, the current approach does not account for mutual occlusion between bubbles, which can impact the accuracy of segmentation and subsequent calculations. Detecting and accurately analysing such complex scenarios involving merging bubbles or mutual occlusion pose challenges that require advanced algorithms or techniques. Future research should focus on developing improved methodologies to handle these limitations and enhance the accuracy of bubble detection and characterization. The coated plate's lower bubble detection accuracy could be attributed to the coating pattern's colour and size. Specifically, the smaller bubbles are similar in size and colour to the coating pattern. Therefore, detecting the bubbles by eyes was challenging, and so the same with the

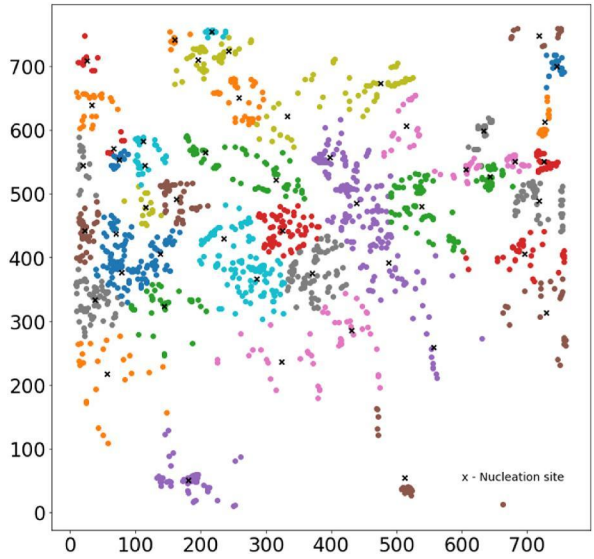


Fig. 16. Evaluation of nucleation site using K-means clustering algorithm, an unsupervised machine learning algorithm.

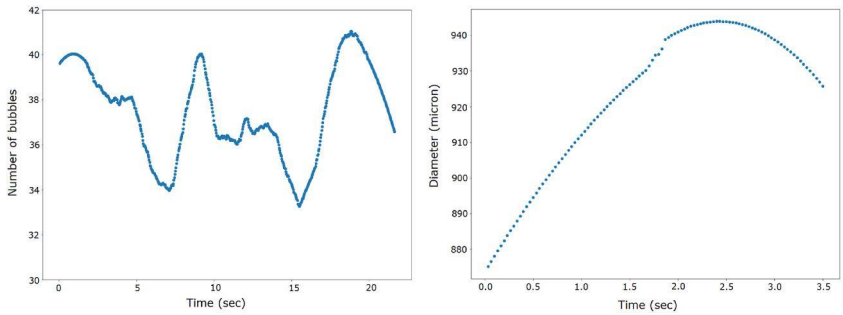


Fig. 17. Number of bubbles variation with time (left), a bubble size variation with time (right) for coated.

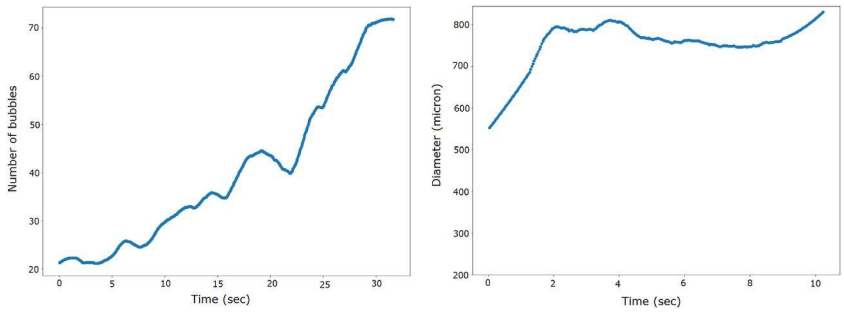


Fig. 18. The number of bubble variations with time (left), a bubble size variation with time (right) for the un-coated plate.

algorithm. Also, more bubbles are visible in the uncoated plate than in the coated plate (Figs. 22 and 23).

4.2. Active nucleation site density

The active nucleation site density is defined as the number of nucleation sites per unit area, N_{ac}, defined as:

$$N_{ac} = \frac{N_s}{A} \ldots\ldots\ldots\ldots \quad (2)$$

where, N_s is the number of nucleation sites, and A is the area of the heated surface. Machine learning was used to calculate the nucleation site, and Fig. 24 shows the detection of nucleation sites for the uncoated and coated plates. The machine learning code

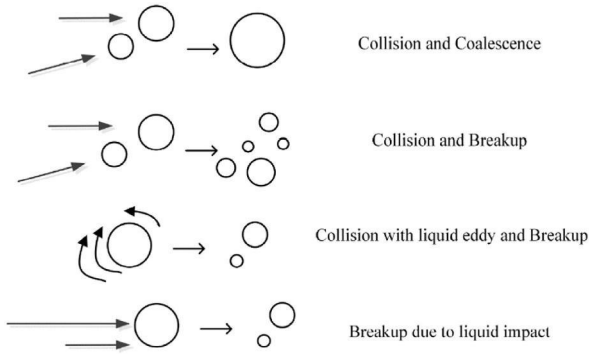


Fig. 19. The phenomenon of bubble collisions results in coalescence and breakup [19].

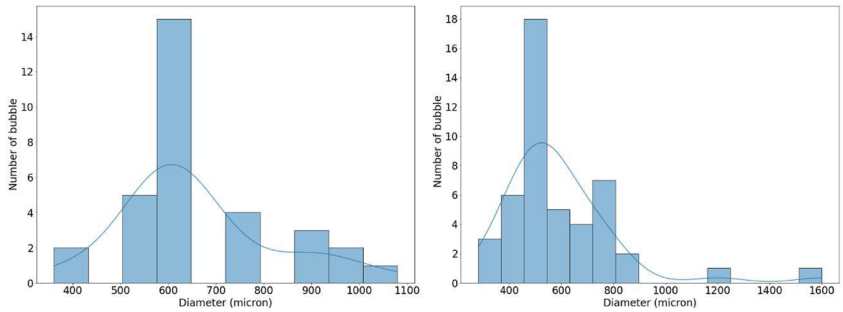


Fig. 20. Log-normal distribution of the bubble diameters histogram (left) for coated plate, (right) for uncoated plate.

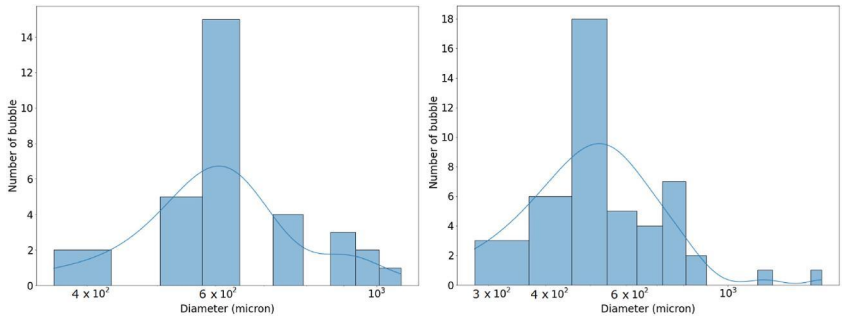


Fig. 21. Normal distribution of the bubble diameters histogram after taking a log in the x scale, (left) for coated plate, (right) for the uncoated plate.

calculates 50 and 42 nucleation sites for the uncoated and coated plates.

The calculated active nucleation site density is N_{ac} , 0.5 $\times$ 10^6 m^{-2} and 0.18 $\times$ 106 m^{-2} for the uncoated and coated plates, respectively.

4.3. Bubble departure diameter

The bubble departure diameter is the diameter at which the bubbles depart from the heating surface. It is calculated as the initial average diameter divided by the number of bubbles [22].

$$d_p = \frac{\sum d_d}{N} \ldots\ldots\ldots \quad (3)$$

The calculated departure diameters using Equation (3) for the coated and uncoated plates are tabulated in Table 1.

4.4. Bubble departure frequency

The departure frequency for each nucleation site can be calculated using Equation 4 [23].

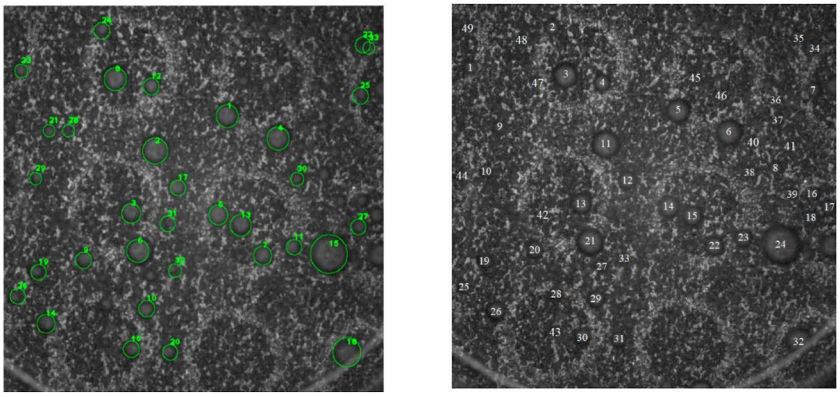


Fig. 22. Bubbles detected by the Deep learning code (left) bubble detection by eye for coated plate (right).

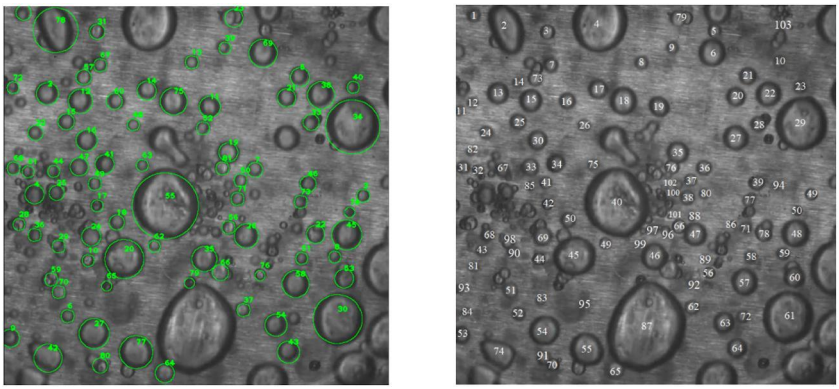


Fig. 23. Bubbles detected by the Deep learning code (left), bubble detection by eye for un-coated plate (right).

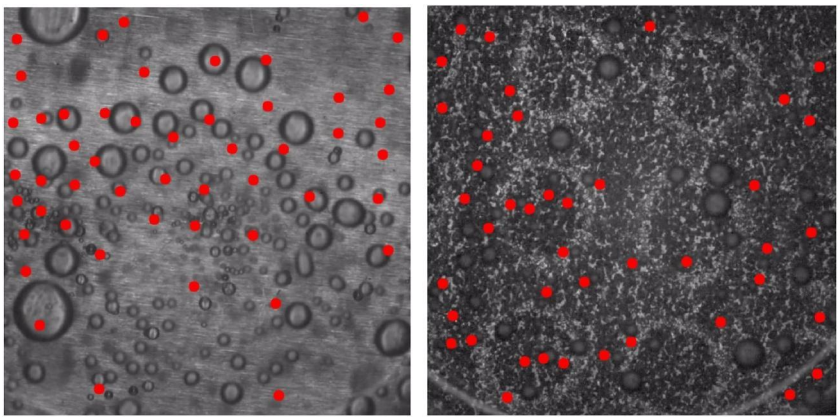


Fig. 24. Nucleation site; uncoated plate (left) and coated plate (right).

$$f_d = \frac{n_{traj} * F_s}{n_{pic}} \ldots\ldots\ldots\ldots \quad (4)$$

where F_s is the frame speed of the high-speed camera and n_{pic} is the number of images used for tracking. n_{traj} is the number of bubble trajectories, and it is obtained by counting the number of the starting coordinates of each trajectory. f_d was calculated using equation (4).

The calculated bubble parameters and the experimental heat transfer performance results for the coated and uncoated plates are shown in Table 3.

Table 3
Bubble parameters results.

Heat exchanger plate	Heat flux, kWm^{-2}	Heat transfer surface area, cm^2	Bubble detection accuracy, %	Nucleation site	Active Nucleation site density, m^{-2}	Average Departure Diameter, μm	Departure frequency, s^{-1}	Heat transfer coefficient, $kWm^{-2}K^{-1}$
Coated	18	10	61.22	42	0.18×10^6	543.26	43	1.25
Uncoated	18	10	78.6	50	0.5×10^6	530.08	53	1.20

An important consideration in our study is the lack of error analysis on the bubble parameters calculated by the algorithm. We had planned to calculate the same set of bubble parameters from the side-view experiment. Unfortunately, we could not obtain the experimental values of these parameters from the videos, mainly due to the unclear and out-of-focus bubbles. The rig's design was not optimal for capturing the side view of the bubbles while maintaining good focus. It ultimately restricted our ability to quantify the accuracy and uncertainty of the measurements by the algorithm.

Table 3 shows that the bubble departure diameter for the uncoated plate is within 3% of the coated plate. The heat transfer performance of these uncoated and coated stainless plates was measured in the GeoHex test rig for the same experimental conditions, and the difference in heat transfer coefficient was only 4%. Therefore, no significant difference has been found between the coated and uncoated plate with respect to the results of bubble departure diameter and heat transfer coefficient. However, we see from Table 1 other bubble parameters; the nucleation site and the departure frequency for the coated plate are 16% and 18% lower than for the uncoated plate. The departure frequency and nucleation site are calculated using the total number of bubbles detected; therefore, they depend on the number of bubbles detected, which is ~17% lower for the coated plate than those detected in the uncoated plate. As seen from all the bubble images for the coated plate, the coated pattern and colour are pretty much like the bubble size and colour. Therefore, the absence of contrast between the coated plate pattern and bubble size and colour could be the reason for lower bubble detection efficiency in the coated plate than in the uncoated plate. However, the coated pattern didn't impact the bubble diameter calculation because the diameter or size was determined for the detected bubbles and did not depend on the number of bubbles detected.

We believe the lower bubble detection efficiency was related to the coating pattern for the coated plate. The lower detection efficiency for the coated plate affects the other bubble parameters except for the bubble diameter, which is within 3%. The results of bubble departure are well agreed with the heat transfer coefficient results for these two plates. Therefore, we can say that there isn't any significant difference between the coated and uncoated plates concerning bubble parameters. The Oleophilic (Hydrophobic dots [Si:Zr:Nd] over hydrophilic [Si:Zr:La] surface coating didn't improve the heat transfer performance.

The bubble parameters, such as bubble departure diameter and departure frequency, were calculated in reference [22,24] for the working fluid R134a for subcooled flow boiling [25]. investigated R134a for pool boiling. However, in references [22,24,25], the heating surface and the experimental conditions differ from those used in this work. Nuclear boiling is very sensitive to the state of the boiling area [25]. Surface characteristics, fluid properties, and operating parameters such as system pressure and temperature significantly influence the nucleation process and bubble parameters [26]; Mahmoud & Karayiannis, 2018; Stojanovic et al., 2022). Surface roughness enhances heat transfer by increasing surface area, which tends to increase the number of nucleation sites, thereby generating a greater number of bubbles [9]. Bubble diameter also depends on the thermophysical properties of refrigerants or working fluid [25]. The parameters of the bubble, such as the departure diameter of the bubble, increase with the contact angle and heat flow rate; the departure frequency of the bubble is also dependent on surface and experimental conditions [12,27]. The nucleation or active nucleation sites increase with surface roughness [28]. The nucleation site density depends not only on the surface physical properties and the surface finish but also on the liquid physical properties and the surface temperature [5]. In nucleate boiling, the heat is transferred through latent heat transfer; in Ref. [25], heat transfer performance was observed for R-410 A due to a higher latent heat value than R-134a under the identical experimental conditions. Other works [26]; Paz et al. (2017), SAKASHITA (2011) analysed nucleate boiling and evaluated the bubbles parameters; however, these works were done for the water or other heat transfer fluids. Considering the heated surface materials, and experimental conditions, the bubble parameters evaluated in this work cannot be compared with references [22,25] since this work has been carried out to produce bubbles on a heated surface with different surface characteristics and operating conditions than that used in the mentioned references. Others worked using different refrigerants or water bubbles; as discussed above, the results in this work cannot be compared.

This study introduces a novel approach by exploring bubble production on a newly developed coated heat exchanger surface. The coating, consisting of oleophilic dots (Si:Nd:Zr-57:26:17 at%) over hydrophilic material (Si:La:Zr-60:20:20 at%) on a stainless steel substrate, was designed through surface engineering to enhance nucleate boiling. These analyses utilise a deep learning algorithm developed in this work to capture and analyse bubble images on the coated surface, comparing the obtained bubble parameters with those from an uncoated bare stainless-steel plate under identical experimental conditions within an Organic Rankine Cycle (ORC) setup. This work showcases the potential of combining deep learning and comparative analysis to investigate the effects of novel surface-engineered coatings on bubble dynamics and heat transfer performance in heat exchanger applications.

5. Conclusions and Recommendations

This study employed deep learning techniques to analyse bubble images of R134a fluid on bare and oleophilic-coated stainless steel heat exchanger plates. The algorithm demonstrated a notable bubble detection accuracy of 61.22% and 78.6% for the coated and

uncoated plates, respectively. According to the calculations, the coated plate yielded a nucleation site count of 42, an active nucleation site density of 0.18×10^6 m^{-2}, an average departure diameter of 543.26 μm, and a departure frequency of 43 bubbles per second. In comparison, the uncoated plate exhibited a nucleation site count of 50, an active nucleation site density of 0.50×10^6 m^{-2}, an average departure diameter of 530.08 μm, and a departure frequency of 53 bubbles per second. The calculations revealed a 3% difference in the departure diameter, closely aligning with the 4% variation in the experimental heat transfer coefficient. However, there is a 17% difference between the coated and uncoated plates in the departure frequency and active nucleation site density. We believe the lower bubble detection efficiency and, therefore, the lower departure frequency and nucleation site for the coated plate were related to the coating pattern. Therefore, it can be concluded that no significant differences regarding the bubble's parameters have been found between the coated and uncoated stainless-steel plates. The Oleophilic (Hydrophobic dots [Si:Zr:Nd] over hydrophilic [Si:Zr:La] surface coating didn't show any improvement concerning the heat transfer performance of the heat exchanger.

In conclusion, our study has provided valuable insights into the analysis of bubble behaviour on heat exchanger plates using deep learning techniques. However, it is important to acknowledge the limitations we have identified. The suboptimal segmentation of very small bubbles and the challenges posed by mutual occlusion may have resulted in some errors when calculating departure diameters and nucleation sites. While the side view offers advantages in accurate parameter calculation, its practicality is limited due to constraints in experimental rig design. Future research should focus on exploring advanced machine learning techniques, such as time-dependent classification algorithms, to overcome these limitations and improve the accuracy of bubble detection as well as the calculation of key parameters. Our study lays the foundation for future research to overcome these challenges and further advance the understanding of bubble dynamics and heat transfer performance in coated and uncoated heat exchanger plates.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:Mahfuza Ahmed reports financial support was provided by European Commission.

Data availability

Data will be made available on request.

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